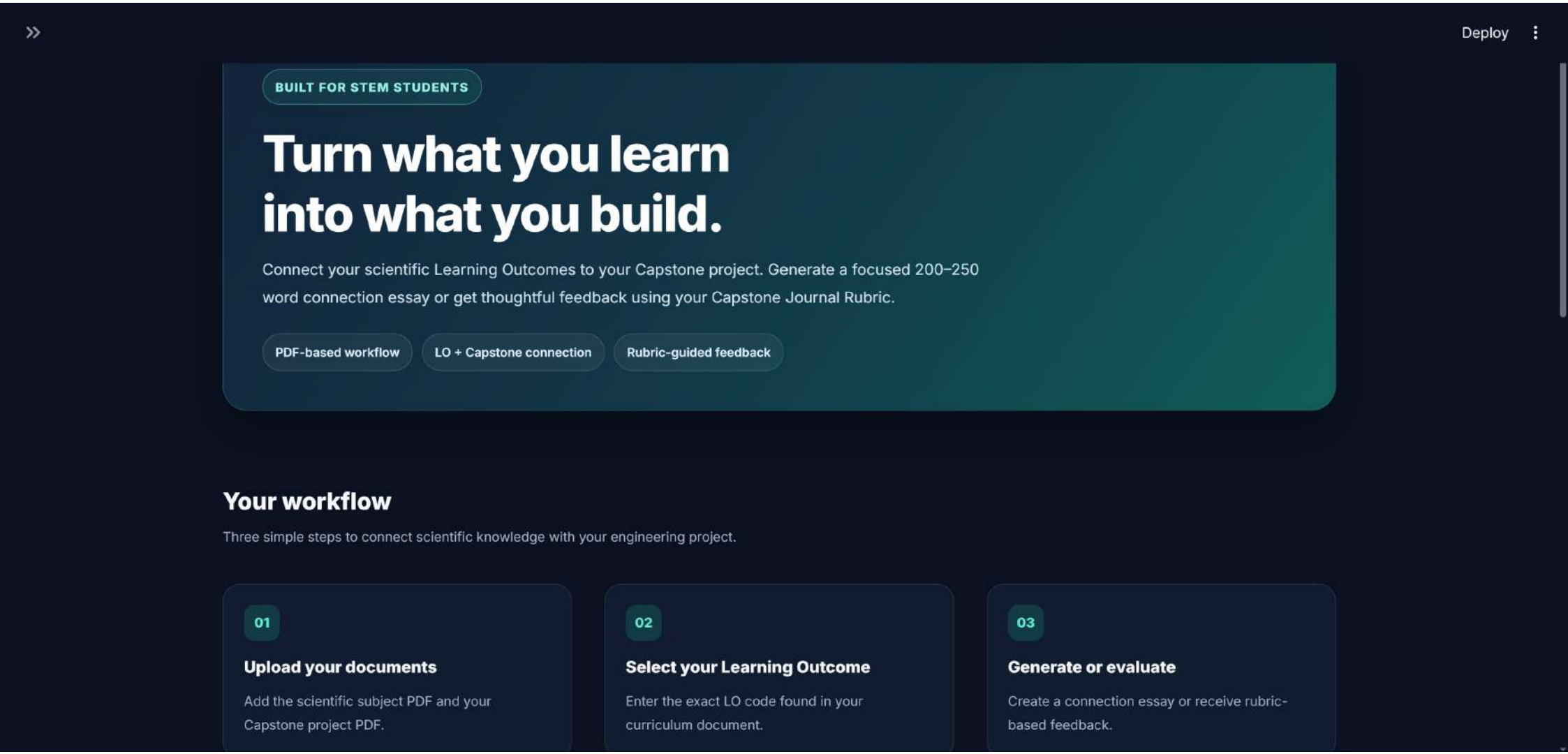

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BUILT FOR STEM STUDENTS
Turn what you learn into what you build.
Connect your scientific Learning Outcomes to your Capstone project. Generate a focused 200–250 word connection essay or get thoughtful feedback using your Capstone Journal Rubric.
PDF-based workflow
LO + Capstone connection
Rubric-guided feedback
Your workflow
Three simple steps to connect scientific knowledge with your engineering project.
01
Upload your documents
Add the scientific subject PDF and your Capstone project PDF.
02
Select your Learning Outcome
Enter the exact LO code found in your curriculum document.
03
Generate or evaluate
Create a connection essay or receive rubric-based feedback.

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Your workspace

Upload the two documents you want STEM Connect AI to analyze. Your files determine the content of the analysis.

DOCUMENT 01

Scientific subject

Curriculum, Learning Outcomes, scientific concepts, and skills.

Upload scientific subject PDF

Upload 200MB per file • PDF

DOCUMENT 02

Capstone project

Project requirements, design choices, evidence, and project progress.

Upload Capstone project PDF

Upload 200MB per file • PDF

What would you like to do?

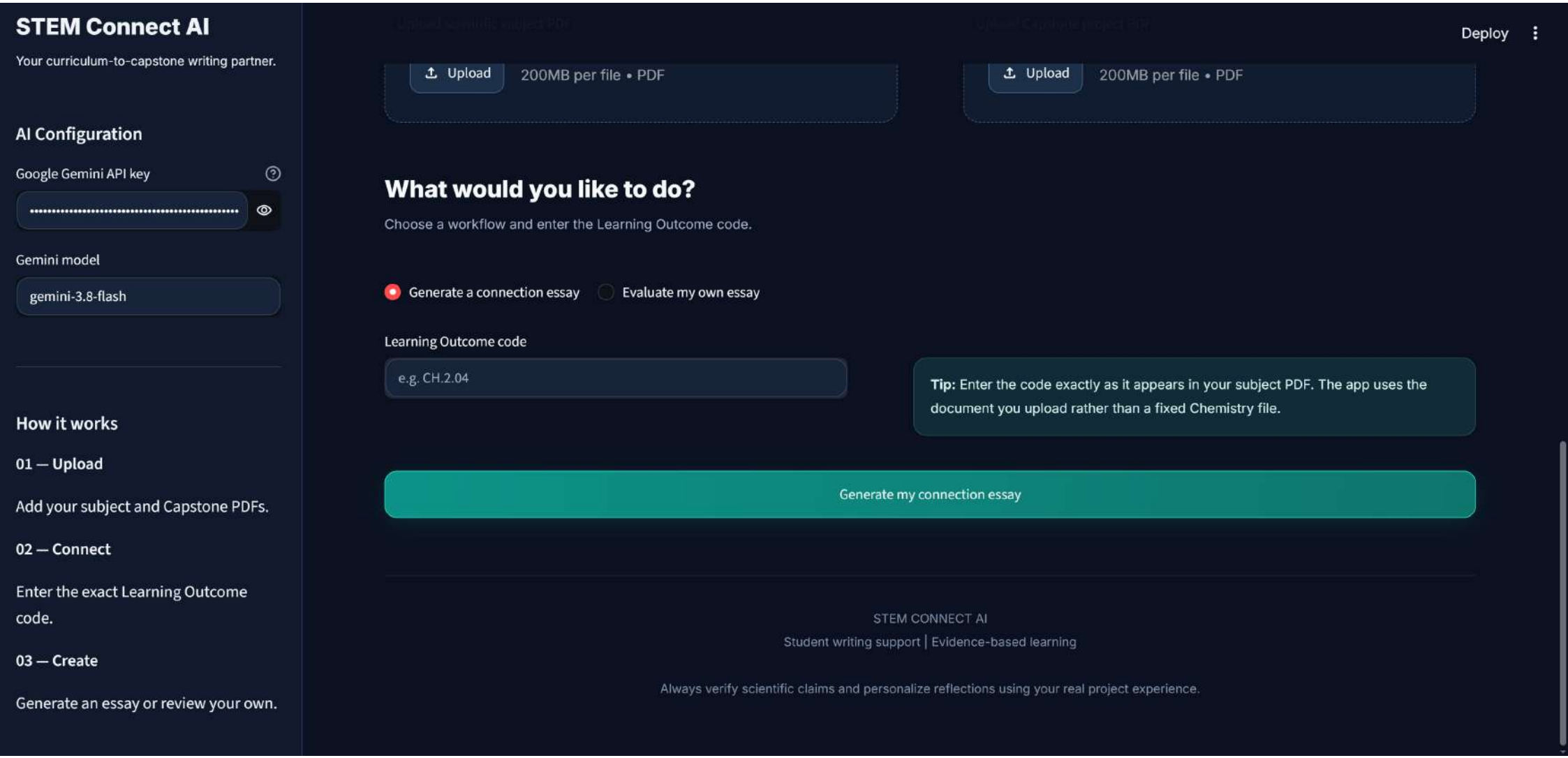
Choose a workflow and enter the Learning Outcome code.

(•) Generate a connection essay ( ) Evaluate my own essay

Learning Outcome code

e.g. CH.2.04

Tip: Enter the code exactly as it appears in your subject PDF. The app uses the document you upload rather than a fixed Chemistry file.


STEM Connect AI
Your curriculum-to-capstone writing partner.
AI Configuration
Google Gemini API key
Gemini model
gemini-3.8-flash
How it works
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Add your subject and Capstone PDFs.
02 — Connect
Enter the exact Learning Outcome code.
03 — Create
Generate an essay or review your own.
Deploy
Upload
200MB per file • PDF
Upload
200MB per file • PDF
What would you like to do?
Choose a workflow and enter the Learning Outcome code.
Generate a connection essay
Evaluate my own essay
Learning Outcome code
e.g. CH.2.04
Tip: Enter the code exactly as it appears in your subject PDF. The app uses the document you upload rather than a fixed Chemistry file.
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Student writing support | Evidence-based learning
Always verify scientific claims and personalize reflections using your real project experience.

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Your essay is ready. Review it and personalize details that only you know.

Your connection essay

Review the generated draft before using it.

GENERATED DRAFT

Investigating Reaction Rates in Smart Food Packaging Systems

Chemical kinetics examines the rate of chemical reaction, describing how rapidly reactants convert into products under specific conditions such as reactant concentration, temperature, and atmospheric exposure. In the Capstone Challenge, this scientific principle directly informs smart food packaging designed to delay food deterioration. The project requires engineering a portable prototype with an internal volume between 1.0 and 3.0 L to slow physical and chemical deterioration by controlling environmental drivers, including oxygen, moisture, or light exposure. The empirical determination of reaction rates can be investigated by comparing fresh food preserved within the prototype against an unpreserved control sample across defined storage intervals: 0, 24, 48, 72, 96, and 120 hours. Collecting quantitative readings of at least two approved measurable changes—such as mass loss, pH shifts, or specific degradation indicators—allows the construction of time-dependent curves that illustrate deterioration progression over a multi-day testing period. A primary scientific limitation of this framework is that tracking macroscopic degradation does not isolate individual chemical reaction pathways or determine formal kinetic parameters, such as reaction orders or rate constants derived from collision theory. Instead, testing measures composite empirical trends in complex biological materials. Despite these kinetic limitations, using systematic time-series measurements and controlled comparative testing highlights the essential role of evidence-based engineering in verifying

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Word count	Evidence pages	Concepts checked
221	4	6
↑ Target: 200–250	↑ Retrieved from Capstone	↑ LO connection review

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Before you submit

Check these details and personalize them using your real project experience.

- Identify the specific fresh food selected and its primary deterioration pathways.
- Specify the two approved quantitative quality indicators monitored during testing.
- Describe the exact natural or repurposed barrier materials fabricated into the prototype.
- Detail the specific natural pigment used for the sensing indicator and its calibrated response scale.

Connection analysis

SUMMARY

The Capstone challenge requires students to slow the rate of fresh food deterioration (which includes chemical deterioration) through packaging and monitor time-dependent changes over 120 hours. While empirical monitoring of deterioration rates over time is supported, formal chemical kinetics concepts such as reaction order, molecularity, and the law of mass action are not mentioned or required in the Capstone documentation.

STRONGEST DEFENSIBLE CONNECTION

The strongest connection is to 'Rate of chemical reaction', where students are explicitly tasked with designing packaging to slow the rate of chemical deterioration of fresh food by altering atmospheric and physical exposure factors.

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Missing information and limitations

- Formal kinetic modeling requirements (e.g., calculation of reaction orders, rate constants, or activation energy).
- Explicit connection between deterioration rates and molecular-level collision theory.
- Guidelines on whether chemical degradation reactions in food should be modeled using specific rate laws or the law of mass action.

Concept-by-concept check

Review which scientific connections are supported by your project evidence.

Order.

Status: UNSUPPORTED

Reaction order refers to the power-law dependence of reaction rate on reactant concentrations in a rate law. The Capstone evidence does not reference reaction orders, rate laws, or mathematical kinetic modeling.

Limitation: The Capstone evidence does not mention reaction order or rate law calculations.

Molecularity.

Status: UNSUPPORTED

Rate of chemical reaction.

Status: SUPPORTED

The Capstone challenge directly addresses controlling and slowing the rate of food deterioration, explicitly mentioning the influence of packaging on the rate of chemical deterioration.

> Food packaging can influence the rate of physical, chemical, and biological deterioration by controlling the environment surrounding food.

Capstone PDF page 1

Limitation: The focus is on overall food deterioration rates rather than isolating a single isolated chemical reaction.

Law of mass action.

Status: UNSUPPORTED

The law of mass action relates reaction rates and chemical equilibria to the active masses or concentrations of reactants. It is not mentioned or applied in the Capstone documentation.

Limitation: No connection to the law of mass action is present in the Capstone evidence.

Determination of rate of reaction.

Status: LIMITED

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What would you like to do?

Choose a workflow and enter the Learning Outcome code.

Generate a connection essay Evaluate my own essay

Learning Outcome code

CH.2.04

Tip: Enter the code exactly as it appears in your subject PDF. The app uses the document you upload rather than a fixed Chemistry file.

Your essay

Paste your own work. The app will evaluate it using the rubric without rewriting your essay.

Chemical kinetics examines the rate of chemical reaction, describing how rapidly reactants convert into products under specific conditions such as reactant concentration, temperature, and atmospheric exposure. In the Capstone Challenge, this scientific principle directly informs smart food packaging designed to delay food deterioration. The project requires engineering a portable prototype with an internal volume between 1.0 and 3.0 L to slow physical and chemical deterioration by controlling environmental drivers, including oxygen, moisture, or light exposure. The empirical determination of reaction rates can be investigated by comparing fresh food preserved within the prototype against an unpreserved control sample across defined storage intervals: 0, 24, 48, 72, 96, and 120 hours. Collecting quantitative readings of at least two approved measurable changes—such as mass loss, pH shifts, or specific degradation indicators—allows the construction of time-dependent curves that illustrate deterioration progression over a multi-day testing period. A primary scientific limitation of this framework is that tracking macroscopic degradation does not isolate individual chemical reaction pathways or determine formal kinetic parameters, such as reaction orders or rate constants derived from collision theory. Instead, testing measures composite empirical trends in complex biological materials. Despite these kinetic limitations, using systematic time-series measurements and controlled comparative testing highlights the essential role of evidence-based engineering in verifying that a functional packaging system reliably mitigates food deterioration.

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Evaluate my essay with the rubric

> Review extracted Learning Outcome

Your rubric-based feedback is ready.

Your rubric feedback

A formative estimate to help you improve. This is not an official grade.

OVERALL ESTIMATED LEVEL

Insufficient evidence

Formative feedback — Not an official grade

The essay provides a strong, scientifically grounded discussion of chemical kinetics and prototype testing methodology within the Capstone Challenge. However, it functions as an objective technical report rather than a complete journal entry, omitting any discussion of team collaboration or personal growth. As two of the four rubric criteria cannot be evaluated from the text, there is insufficient evidence to determine an overall level.

Deploy

Criteria breakdown

Each criterion is evaluated separately against the supplied Capstone Journal Rubric.

Learning Transfer

GREEN

Evidence from your essay

Chemical kinetics examines the rate of chemical reaction, describing how rapidly reactants convert into products under specific conditions such as reactant concentration, temperature, and atmospheric exposure. In the Capstone Challenge, this scientific principle directly informs smart food packaging designed to delay food deterioration.

Why this level?

The response demonstrates clear evidence of synthesizing chemical kinetics concepts with the practical objectives of the Capstone project, including environmental controls and degradation tracking.

How to improve

To reach the Blue level, extend your synthesis beyond testing metrics to reflect on how principles learned across the entire project (such as material selection or prototyping iterations) connected with this kinetic framework.

Question to think about

How did your understanding of chemical kinetics shape the initial physical design choices you made for the packaging prototype?

Team Collaboration

Insufficient evidence

Evidence from your essay

None provided in the text.

Why this level?

The essay contains no mention of working in a team, individual responsibilities, or collaborative processes.

How to improve

Include a discussion of your specific role on the team and detail how collaborative behaviors or division of labor impacted the project.

Question to think about

What specific role did you play in planning and carrying out the testing protocols alongside your team members?

Using the Engineering Design Process

GREEN

Evidence from your essay

The empirical determination of reaction rates can be investigated by comparing fresh food preserved within the prototype against an unpreserved control sample across defined storage intervals... A primary scientific limitation of this framework is that tracking macroscopic degradation does not isolate individual chemical reaction pathways... controlled comparative testing highlights the essential role of evidence-based engineering...

Deploy

Using the Engineering Design Process

GREEN

Evidence from your essay

The empirical determination of reaction rates can be investigated by comparing fresh food preserved within the prototype against an unpreserved control sample across defined storage intervals... A primary scientific limitation of this framework is that tracking macroscopic degradation does not isolate individual chemical reaction pathways... controlled comparative testing highlights the essential role of evidence-based engineering...

Why this level?

The response provides a clear and adequately developed reflection on the testing and evaluation stage of the engineering design process, thoughtfully identifying limitations of macroscopic data collection.

How to improve

Explain how the insights and limitations identified during testing directly informed your decisions to redesign, iterate, or refine the prototype.

Question to think about

How could the scientific limitations you identified lead to concrete modifications in your testing protocol or prototype design?

Personal Reflection

Insufficient evidence

Evidence from your essay

None provided in the text.

Deploy

The text is written in an impersonal, third-person format and does not share viewpoints, interpretations, or examples regarding personal growth.

How to improve

Share first-person reflections detailing personal obstacles you encountered, how you overcame them, and how your engineering perspective evolved.

Question to think about

What was the most challenging realization you had while evaluating these experimental limitations, and how did it influence your personal growth as an engineer?

Your next steps

- Add a dedicated section detailing how your team functioned, outlining your individual responsibilities and how the group resolved challenges.
- Incorporate personal reflection identifying your own learning curve, challenges faced during testing, and specific areas of growth.
- Connect the testing limitations directly back to engineering design iterations to demonstrate the full cycle of the engineering design process.

The overall estimated level is formative feedback to guide revision and reflection, not an official grade.

Download rubric feedback as TXT